CHD classification with support vector machines

SAheart data | 462 observations | 9 predictors

> **Selected: linear SVM, C=0.1.** CV error 28.8% ± 1.4% SE. It is the simplest model within one SE of the minimum CV error.

Data and model selection

There are 302 non-cases and 160 CHD cases (34.6%); no missing values; 83 observations have at least one 1.5-IQR predictor outlier. SVM comparison is reasonable for this moderate-sized binary task. Famhist was coded Absent=1, Present=2; scaling was fit within training folds. A stratified 25% test set (seed 432) was reserved before tuning; training used 5-fold CV repeated 3 times. The 67 candidates included linear, RBF, and degree-2/3 polynomial kernels (polynomial coef0=1); C={0.01, 0.1, 1, 10, 100}; gamma={0.001, 0.01, 0.1, 1}, plus two quadratic boundary probes: (C=0.001, gamma=1) and (C=0.01, gamma=10).

Kernel family	Best parameters	Mean CV error ± SE	Support vectors	Selection
Linear	C=0.1	28.8% ± 1.4%	216	CHOSEN
Polynomial (degree 2)	C=0.01, degree=2, gamma=1, coef0=1	28.1% ± 1.3%	209	
Polynomial (degree 3)	C=10, degree=3, gamma=0.01, coef0=1	27.4% ± 1.5%	207	
RBF	C=10, gamma=0.01	27.2% ± 1.7%	210	

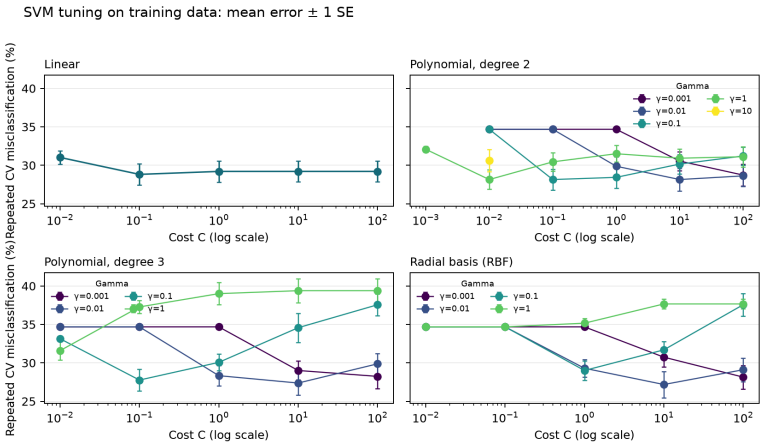


Figure 1. Training CV error ± 1 SE; color denotes gamma, horizontal axis is log(C). Test data were not used for tuning.

Held-out test performance

Test error: **25.9% (30/116)**
Confusion matrix (actual rows 0,1; predicted columns 0,1):
[[65, 11], [19, 21]]
FP=11; FN=19

Sensitivity=52.5%; specificity=85.5%. For screening, false negatives may be costlier because follow-up can be delayed; false positives can prompt unnecessary assessment.

The selected fit uses 216 support vectors (62.4% of training cases) and has a linear boundary in standardized predictor space. RBF had the lowest CV error (27.2% ± 1.7% SE), but its advantage over the selected linear model was not clear by the one-SE rule. This is predictive, not causal; individual predictor effects are not claimed.

All test metrics are from the single held-out evaluation. Full per-setting results and reproducible code accompany this report.